PRN No.	Total No. of Questions: 06

QP Code:

JSPM's
Rajarshi Shahu College of Engineering, Tathawade, Pune- 411033
(An autonomous institute affiliated to Savitribai Phule Pune University)

Examination: Mid Semester Examinations (MSE)

Semester: II **Academic Year: 2023-24**

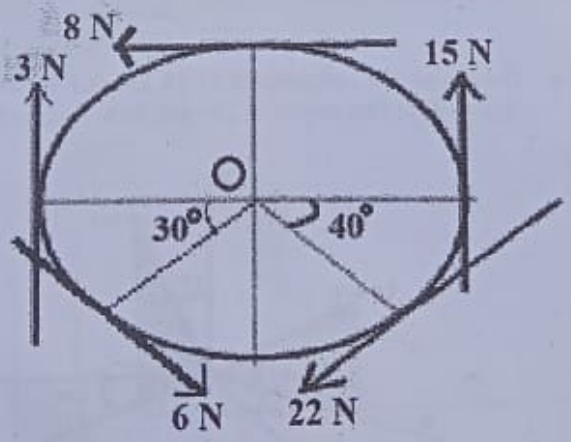
Class: F. Y. B. Tech. (All Programs)
Department: Engineering Sciences and Humanities

Subject Code: CE1201 **Subject Name and pattern: ENGINEERING MECHANICS 2023**

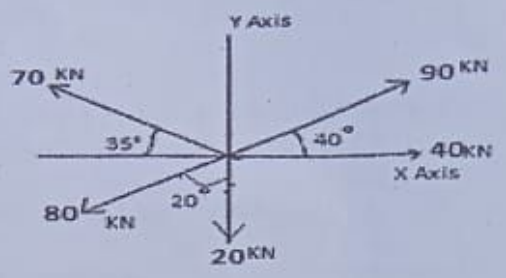
Duration: 1 Hour 15 Minutes **Max. Marks: 30 Marks**

Instructions to the Candidates
1. Solve Q.1or Q.2, Q.3 or Q.4 and Q.5 or Q.6.
2. Assume suitable and necessary data wherever required.
3. Neat diagram must be drawn wherever necessary.

Q. No.			Marks	BL	CO
1	a	Determine the resultant of forces tangent to the circle of radius 3 m as shown in below figure. Determine its location with respect to point 'O'. Also find out direction of resultant force.	5	BL2	CO1

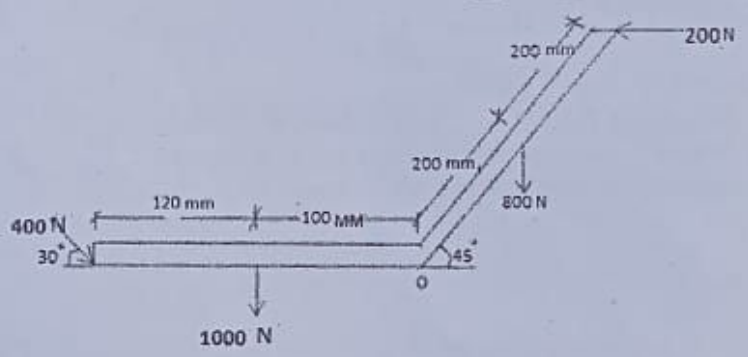


	b	Determine the magnitude and direction of resultant force for following force system.	5	BL2	CO1



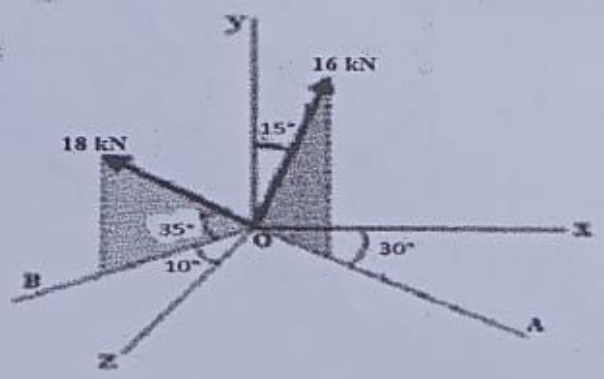
OR

2 a A system of force acting on a bell crank as shown. Determine the magnitude, direction and point of application of resultant with respect to point 'O'. 5 BL2 CO1

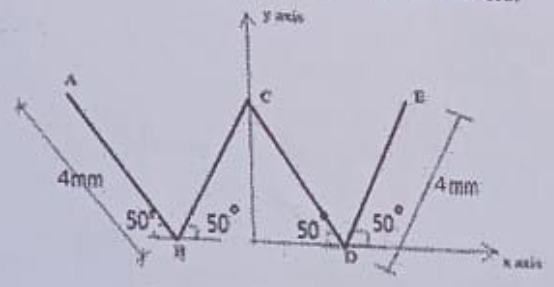


b The sum of two concurrent forces P & Q is 300 N and their resultant is of 150 N. If resultant is perpendicular to P. Find P and Q. 5 BL2 CO1

3 a Determine the components of 18 kN and 16 kN forces as shown in figure. Also find angles θx , θy and θz with co-ordinate axis. 5 BL2 CO2

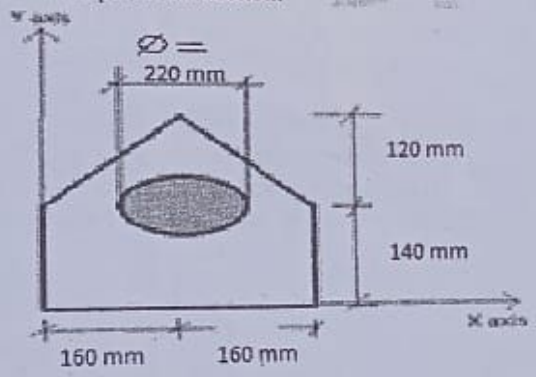


b Determine the centroid of thin wire bend in the shape of 'W' as shown in below figure having 4mm length of all members. 5 BL2 CO2

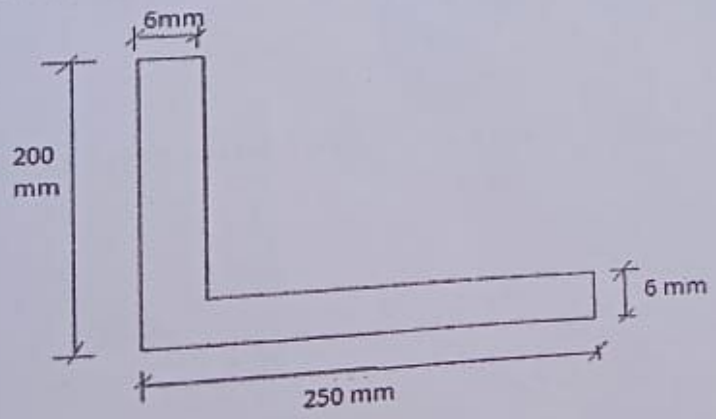


OR

4 a Determine the centroid of following composite section by removing the shaded circular part from section. 5 BL2 CO2



b Find out the moment of inertia of L section shown in fig. about Horizontal Centroidal axis Ixx. 5 BL3 CO2



5 a A small part in a mechanism travel on straight line such that its position $X=t^4-12t^3+20$, where X is in 'meter' and time is in 'second'.

Determine -

a) Time when velocity is zero (v=0) 5 BL2 CO

b) Time when acceleration is zero (a=0)

c) Minimum speed reached by particle.

d) Distance travelled in 4 second

b A projectile is fired with an initial velocity 240 m/s at a target B, located 600 m above the gun and at a horizontal distance of 3600 m. Neglecting air resistance determine the value of the firing angle. 5 BL2 C

OR

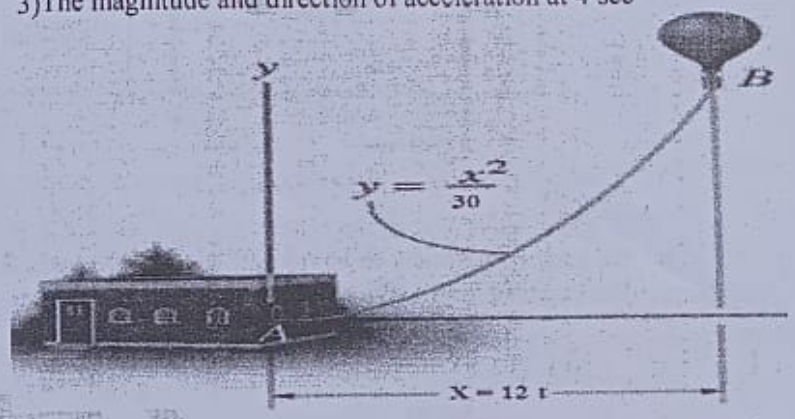
6 a A stone is thrown up with velocity of 30 m/s from ground. On return journey, it strikes a glass placed at half the height & it loses 25% of its velocity in breaking the glass. Find the velocity with which it will strike the ground. 5 BL2 C

b At any instant horizontal position of balloon is shown determine-

1)The distance of the balloon from A at 4 sec

2)The magnitude and direction of velocity at 4 sec

3)The magnitude and direction of acceleration at 4 sec



5 BL2

**********BEST of LUCK**********